**WEEK 11 QUESTION BANK**

1. What is Natural Language Processing (NLP)? Explain how NLP is used to make computers understand human languages.
2. Differentiate between Rule-Based and Statistical approaches in NLP with suitable examples.
3. Explain any five real-world applications of Natural Language Processing.
4. Explain any four commonly used NLP tools and libraries and describe their applications.
5. Differentiate between sentence tokenization and word tokenization with suitable examples.
6. Differentiate between stemming and lemmatization with suitable examples.
7. What is vectorization in NLP? Explain how text data is converted into numerical vectors with a suitable example.
8. Differentiate between One-Hot Encoding and Count Vectorizer with suitable examples.
9. N-grams are defined as the combinations of N Keywords together consider the given "The greatest glory in Living lies not in never falling but in raising every Lies"

   Generate bi grams for the above

   Generate Tri grams for the above

10. Through a step-by-step process calculate TF/IDF for the given corpus and mention the words having highest value

    Doc1: we are going to Mysore
    Doc2: Mysore is a famous place
    Doc3: we are going to famous place

11. Implement CountVectorizer and construct a Document-Term Matrix for three given sentences.
    a. 'He is a smart boy. She is also smart', 'Chirag is a smart person'
    b. 'coronavirus is a highly infectious disease', 'coronavirus affects older people the most', 'older people are at high risk due to this disease'.
    c. """Natural language processing refers to the branch of computer science concerned with giving computers the ability to understand text"""